

GPS CLOCK Using Arduino

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Global positioning system (GPS) synchronised clocks give accurate time. These clocks are universal and commonly used at rail-

way stations, bus stands and airports. These are widely used for military purposes too.

Here we describe a GPS clock based on Arduino Uno R3—an AVR ATmega328-based microcontroller board with six analogue inputs pins and 14 digital input/output (I/O) pins. The microcontroller has 32kB ISP flash

memory, 2kB RAM and 1kB EEPROM. The board provides serial communication via UART, SPI and I2C.

Circuit and working

Figs 1 and 2 show block diagram and circuit of the GPS clock using Arduino, respectively. In addition to Arduino Uno board (BOARD1), the circuit uses SIM28M GPS receiver module (GPS1) along with a GPS antenna (ANT.1), 9V DC power supply adaptor and a few jumpers for header connections.

Here, we have tried to extract GPS time and date from a string (\$GPRMC) that comes from GPS. This string is about 70 characters long. As shown in Fig. 2, Arduino controls all the processes and receives GPS output signals. After receiving the GPS output, Arduino reads all the strings and stores required strings in a string or in an array in the Arduino program. After storing the required string, Arduino extracts the time and date from the stored string and sends it to LCD1 for displaying time and date.

Data pins D4 through D7 of LCD1 are directly connected to pins 5, 4, 3 and 2 of Arduino, respectively. Control pins EN and RS of LCD1 are connected to pins 11 and 12 of Arduino, respectively. GPS receiver's Tx pin is directly connected to Rx pin of Arduino board. Note that the ground pins of Arduino and GPS should be connected to each other.

Here GPS module is operated on 9600bps baud rate. Arduino is configured at 9600bps baud rate by using class Serial.

PARTS LIST

Semiconductors:

- BOARD1 - Arduino Uno board
- LCD1 - 16×2 LCD
- GPS1 - SIM28M GPS receiver with antenna

Resistors (all 1/4-watt, ±5% carbon):

- R1 - 1-kilo-ohm
- R2 - 470-ohm

Miscellaneous:

- 230V AC to 9V DC, 1A adaptor×2
- Jumper wires
- GPS antenna (ANT.1)

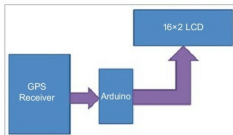


Fig. 1: Block diagram of GPS clock using Arduino

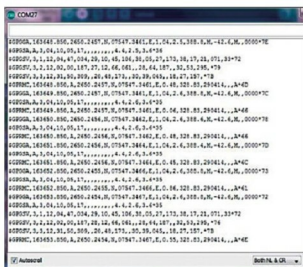


Fig. 3: GPS output from serial monitor of Arduino IDE

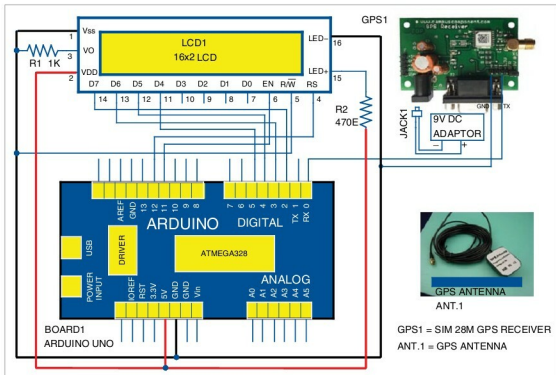


Fig. 2: Circuit diagram of GPS clock using Arduino